

Firmware 7.15 · Block 6 audit report

Ripple memo, displayed address and XRPL length prefix · 22 September 2026

Local audit judgment: selected P06 behavior passes the pinned current-source review and executed tests. P06-F001 is a separate unreachable generic helper-domain edge, recorded for follow-up. Copilot review is a separate final checkpoint, not yet claimed in this report.

Identity and canonical method

Review PR [#853](#). Base: audit/715-p04-final-review @ 92931c91dc77bf2cdf80aaa8938b978add0cfde8; report-generation predecessor: 1444ea18c8dfdb61c50bed92b1b2666c4b1baa27. The containing final PR head is recorded in the PR body. Candidate firmware source: audit/715-scope-repair @ 614425a2a14d0113de251e7944118e47f31f5265; canonical product release/7.15 @ af979cd5099349ba58ec10d1b7eedeb67662c602 remains separate. Block 5 was not audited by this unit.

Main-worktree master template SHA256 at report generation:

dc484ea3bb7810b410c43830c307f032ad19276f81c7c6c123c407405d4b27e8. Its progress ledger can change; the numbered audit form matches the committed snapshot with SHA256

f1eb2a6015d73b552aac1406780ff93553530bbbda084a9a46ce5e2e68525d78. The live full-file hash at review request is in the PR body. Historical counts use `git show --format= --numstat --no-renames COMMIT_SHA`; PR counts use `immutable git diff --numstat`

92931c91dc77bf2cdf80aaa8938b978add0cfde8..1444ea18c8dfdb61c50bed92b1b2666c4b1baa27. The containing final head must preserve the same file inventory and counts. Historical counts include documentation, tests and submodule pointer lines; repeated edits can overlap and are not a net product diff. Binary artifacts appear as -.

Review PR file inventory

File	Added	Deleted	Meaning
docs/release/audit-units/P06-audit-report-20260922.html	+18	-0	Audit report/evidence; no firmware behavior
docs/release/audit-units/P06-audit-report-20260922.pdf	-	-	Audit report/evidence; no firmware behavior
docs/release/audit-units/P06-block-scope-20260922.md	+141	-0	Audit report/evidence; no firmware behavior
docs/release/audit-units/P06-ripple-show-address-20260922.png	-	-	Audit report/evidence; no firmware behavior

Exact candidate dependency pins

Submodule	Commit
code-signing-keys	a6470bd8598e5e9a7bfc38bf139a5e5a616f05ec
deps/crypto/trezor-firmware	8a392f70a5d5575ece3dfb35f115d4a4b27f497c
deps/device-protocol	27d3fa1f6215139cde6411f9a2882f36bb373fc9
deps/googletest	7888184f28509dba839e3683409443e0b5bb8948
deps/python-keepkey	b76ee610dd18934ee3aeeb36cfc541e8799eb46d
deps/qrenc/QR-Code-generator	6dfbfdad5d9303ed190d1c3cb7bec34b565b6ce8

Submodule	Commit
deps/sca-hardening/SecAESSTM32	71d356a1141624994cf613bd2d2583892e8e6d5a

Original P06 units

Commit	Change	Files	Added	Deleted
7a8a873c366c1c7844b2a414a4fa187c5617cba5	P06-001 Host memo assertion	2	+22	-1
885609fbe2ddf66b5224f76880d2c8ff3b90eec1	P06-002 Displayed address response	2	+27	-4
cecf1ea20239d140cb43dc0629219fd9179ae452	P06-003 Permanent host assertion	2	+16	-1
39503dacf595821b2b8a1c65e3f44fa9b6352a88	P06-004 Memo length boundary	3	+37	-1

P06-001 Host memo assertion 7a8a873c366c1c7844b2a414a4fa187c5617cba5

The Ripple memo host assertion was unconditionally skipped under a stale “no memo” assumption. The Python pin removed that skip while retaining the full-feature and 7.15.0 gates. Its historical pointer has since been superseded; the current b76ee610d pin still contains the memo/no-memo assertions. The earlier full host JUnit and an owned candidate emulator run pass the test; Bitcoin-only skips Ripple by policy.

File	Added	Deleted
deps/python-keepkey	+1	-1
docs/release/audit-units/P06-001-ripple-host-coverage.md	+21	-0

P06-002 Displayed address response 885609fbe2ddf66b5224f76880d2c8ff3b90eec1

DebugLink screenshot requests could reuse the response arena while address confirmation was pending, leaving the returned RippleAddress empty. The handler now keeps the derived address in a local buffer and writes the shared response after confirmation; cancellation clears the node. An owned pinned emulator screenshot run passes the current address assertion and produced the screen in this report. Physical OLED remains a release gate.

File	Added	Deleted
docs/release/audit-units/P06-002-ripple-display-response.md	+23	-0
lib/firmware/fsm_msg_ripple.h	+4	-4

P06-003 Permanent host assertion cecf1ea20239d140cb43dc0629219fd9179ae452

The displayed-address host test previously tolerated an empty return value. Its pinned update asserted the known public address for the fixed firmware. The current canonical Python pin retains the assertion for 7.14.2 and later; an owned screenshot run and earlier full CI host JUnit execute and pass it. The historical submodule pointer is superseded.

File	Added	Deleted
deps/python-keepkey	+1	-1

File	Added	Deleted
docs/release/audit-units/P06-003-ripple-display-test.md	+15	-0

P06-004 Memo length boundary 39503dacf595821b2b8a1c65e3f44fa9b6352a88

The serializer used two bytes for a 192-byte memo length, shifting signed XRPL transaction bytes. It now uses one byte through 192 and two from 193. Native vectors cover 191, 192, 193 and 199; the full candidate xunit log executes and passes the boundary. The current memo field can reach 192; Bitcoin-only excludes Ripple.

File	Added	Deleted
docs/release/audit-units/P06-004-ripple-length-prefix.md	+19	-0
lib/firmware/ripple.c	+1	-1
unittests/firmware/ripple.cpp	+17	-0

Later current-candidate interaction

Commit	Change	Files	Added	Deleted
6e79e57b7	Later Ripple amount bound	12	+235	-8

Later Ripple amount bound 6e79e57b7

A subsequent release-audit commit raised the Ripple amount serializer bound to the protocol maximum and added a native regression. It changes the same Ripple source file but leaves the 192-byte length condition and displayed-address response ordering intact; the full candidate suite executes the added amount test.

File	Added	Deleted
include/keepkey/firmware/fsm.h	+1	-0
include/keepkey/firmware/home_sm.h	+2	-0
include/keepkey/firmware/ripple.h	+8	-2
lib/board/confirm_sm.c	+5	-1
lib/firmware/fsm.c	+20	-0
lib/firmware/home_sm.c	+37	-1
lib/firmware/mayachain.c	+8	-1
lib/firmware/ripple.c	+1	-1
lib/firmware/storage.c	+8	-2
unittests/firmware/fsm.cpp	+119	-0
unittests/firmware/mayachain.cpp	+6	-0
unittests/firmware/ripple.cpp	+20	-0

Current source and finding disposition

fsm_msgRippleGetAddress holds the derived public address in local storage through confirmation, then fills the shared response. The current Python test asserts that returned address when display is requested; the current

memo test requires full 7.15, checks the Memos tail and a plain send without Memos. `ripple_serializeVarint` uses `<= 192`. The [XRPL Binary Format](#) specifies one prefix byte for 0–192 and two for 193–12480.

P06-F001 — accepted scope limit / follow-up: the generic helper uses `< 918744` where the XRPL maximum is inclusive, and its debug assertion rejects an exactly three-byte remaining buffer. No current 7.15 Ripple call site reaches that length: memo content is at most 199 bytes; address, public key and signature are shorter. This is a concrete broader-helper issue, not a fixed or waived result of P06-004. Physical OLED and full Ripple helper-domain audit remain open follow-up gates. No new defect was identified in the selected reachable P06 behavior.

Executed evidence and limits

Owned full candidate emulator image

sha256:18f7375986270c6bdcea6155e9fe8382d5a4faca8f0466ff3bf727b759a2e9f3 ran `test_ripple_show_address` with `KEEPKEY_SCREENSHOT=1` and exact selection: 1 passed, 2 deselected, one PNG. JUnit SHA256 3c2897b104ba4d8744379d810bb49114993cdc3a14344c31e6ca39f1bfd3ff4b; visually inspected PNG SHA256 8ab53a8b93f7020b1e14f43a943950f27618e10b2d21a6f3ce4c2404f6802a6d. The same image ran `test_sign_with_thorchain_memo`: 1 passed, 3 deselected; JUnit SHA256 73d618dfdf7a2e68d9d0350f9fa9a654525820ac16ed6eabc86b1711d205ff3. Complete local evidence: `/private/tmp/kk-fw-715-p06-evidence`.

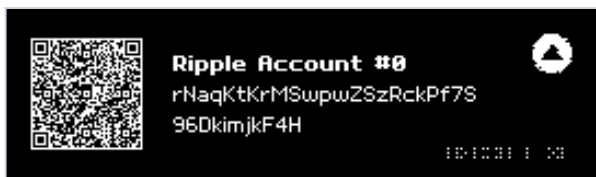
Full candidate make `xunit` executes five Ripple native cases including `Ripple.MemoLengthPrefixBoundary: 573` firmware, 17 board, 18 crypto and 6 Pallas tests passed; log SHA256

5a452156d0b2264f480ac5d640e388e1ea9ab10bf57783751bcde2f947459d. Bitcoin-only deliberately excludes Ripple and passed its 131 firmware, 17 board and 18 crypto tests; log SHA256 4509a2ef1d05d5cdfa21401a6350fd562a48b1b6e69be3dcb14f211ee25ff49e.

Earlier firmware-equivalent [CI run 34951131013](#) at 7e091af157131897adc5514de392570c32088946: full host artifact 10390030970 passes Ripple address/display/sign/memo; Bitcoin-only host artifact 10389023825 skips all seven Ripple cases by design. Native artifacts 10388679573/10389512420 and ARM firmware artifacts 10389287571/10388589162. Full ARM job 104322245689 and Bitcoin-only job 104322245632 both passed cross-compile and SRAM budget steps. From CI source to candidate, only CI workflows changed. From owned-image source to review branch, only audit documentation/SOP changed. These are code/test/pin-equivalent results, not exact-head workflow certification.

Captured emulator screen

The PNG below is committed beside this report and was visually inspected. It is emulator evidence of the displayed public address, not a physical OLED sign-off.



Remaining gates: physical OLED review, signed hardware upgrade, final exact-workflow CI, Block 5, and the broader Ripple helper-domain finding are not completed by this P06 receipt.